4339 Asset Pricing and Investments - Assignment 3

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1 Factors and Arbitrage

We assume that stock returns follow a factor structure with two factors f_1 and f_2. Furthermore, the risk-free rate $r_f = 3\%$, and A and B are two well-diversified portfolios with the following properties:

Portfolio	β_1	β_2	$E[r]$
A	1	0.5	7.5%
B	0.2	2	13.4%

1.1 (a) The price of risk for each factor, f_1 and f_2, under no arbitrage:

For a well-diversified portfolio idiosyncratic risk is diversified away, so expected excess returns are fully explained by the factor exposure,

$$E[r_i] - r_f = \lambda_1 \beta_{i1} + \lambda_2 \beta_{i2},$$

where λ_n is the excess return per unit of exposure to factor n.

Substituting from the initial table yields the following two equations:

$$\begin{aligned} A: &\quad 0.075 - 0.03 = \lambda_1 \cdot 1 + \lambda_2 \cdot 0.5 \\ B: &\quad 0.134 - 0.03 = \lambda_1 \cdot 0.2 + \lambda_2 \cdot 2 \end{aligned}$$

We find the price of risk for each factor by solving equation A for λ_1 and substituting into B:

$$\begin{aligned} A: &\quad \lambda_1 = 0.045 - 0.5\lambda_2 \\ B: &\quad 0.104 = (0.045 - 0.5\lambda_2) \cdot 0.2 + \lambda_2 \cdot 2 \\ &\quad 0.104 = 0.009 + 1.9\lambda_2 \\ &\quad \lambda_2 = 0.05 \quad \Longrightarrow \quad \lambda_1 = 0.02 \end{aligned}$$

The price of risk for factor f_1 is $\lambda_1 = 2.0\%$ and for factor f_2 it is $\lambda_2 = 5.0\%$, in this system.

1.2 (b) Identifying and exploiting an arbitrage opportunity:

C is a well-diversified portfolio with factor loadings $(\beta_{C1}, \beta_{C2}) = (0.5,\ 1.0)$ and expected return $E^{\text{obs}}[r_C] = 9.5\%$:

Portfolio	β_{C1}	β_{C2}	$E^{\text{obs}}[r]$
C	0.5	1.0	9.5%

According to C's factor loadings and the prices of risk calculated in (a), the expected no-arbitrage return of C is

$$\begin{aligned} E^{\text{model}}[r_C] &= 0.03 + \lambda_1 \cdot 0.5 + \lambda_2 \cdot 1 \\ &= 9.0\% \end{aligned}$$

Because the value does not coincide with the observed return in the table, there exists an arbitrage opportunity.

To take advantage of it, we create a portfolio H made up of A, B and the risk-free asset, which replicates the factor exposure of C:

$$\begin{aligned} & 1 = w_A + w_B + w_f \\ f_1: \quad & 0.5 = w_A + 0.2\, w_B + 0\, w_f \\ f_2: \quad & 1 = 0.5\, w_A + 2\, w_B + 0\, w_f \end{aligned}$$

Solving equation f_1 for the weight in portfolio B and substituting into equation f_2 yields the respective weights in portfolio H:

$$\begin{aligned} f_1: \quad & w_B = 2.5 - 5w_A \\ f_2: \quad & 1 = 0.5w_A + 2 \cdot (2.5 - 5w_A) \\ & 1 = 5 - 9.5w_A \\ & 4 = 9.5\, w_A \\ & w_A = \tfrac{16}{38} \quad \Longrightarrow \quad w_B = \tfrac{15}{38} \quad \Longrightarrow \quad w_f = \tfrac{7}{38} \end{aligned}$$

The return of this newly created portfolio H is

$$\begin{aligned} E[r_H] &= w_A\, E[r_A] + w_B\, E[r_B] + w_f\, r_f \\ E[r_H] &= \tfrac{16}{38}(7.5\%) + \tfrac{15}{38}(13.4\%) + \tfrac{7}{38}(3\%) = 9.0\% \end{aligned}$$

As a consistency check, $E[r_H] = 9.0\% = r_f + 0.5\lambda_1 + 1.0\lambda_2 = E^{\text{model}}[r_C]$, which confirms that H replicates C's factor exposures.

Therefore, we would take a long position in portfolio C and an equally sized short position in H. The net investment is zero, the net exposure to each factor is zero, and as such the position earns a riskless profit of $9.5\% - 9.0\% = 0.5\%$ per unit of capital.

2 Performance measures

Consider the following properties for two portfolios:

Portfolio	$E[r^e]$	σ
Market (M)	10%	25%
P	15%	35%

The Sharpe ratios of the two portfolios are

$$S_M = \frac{E[r_M^e]}{\sigma_M} = \frac{0.10}{0.25} = 0.4000, \qquad S_P = \frac{E[r_P^e]}{\sigma_P} = \frac{0.15}{0.35} \approx 0.4286.$$

2.1 (a) Constructing P^*

In the $(\sigma, E[r])$ plane, all weight allocations between a risky portfolio and the risk-free asset lie on a line through $(0, r_f)$, with the slope representing that portfolio's Sharpe ratio. By moving along this linear function, we can adjust $(\sigma, E[r])$, while the Sharpe ratio of P remains constant. In this example, σ_P needs to be lowered to match σ_M. We therefore move left, since $\sigma_M < \sigma_P$.

Let the investor hold a weight w_P in portfolio P and $w_f = 1 - w_P$ in the risk-free asset. The excess return of the combined portfolio is

$$\begin{aligned} E[r_{P^*}] &= w_P\, E[r_P] + (1 - w_P) r_f \\ E[r_{P^*}^e] &= w_P\, E[r_P^e] \end{aligned}$$

since the risk-free asset contributes no excess return. The portfolio volatility is

$$\begin{aligned} \sigma_{P^*} &= \sqrt{w_P^2 \sigma_P^2 + w_f^2 \sigma_f^2 + 2 w_P w_f \sigma_{P,f}} \\ \sigma_{P^*} &= \sqrt{w_P^2 \sigma_P^2} = w_P\, \sigma_P, \end{aligned}$$

since the risk-free asset has zero variance and zero covariance with P.

We see that both $E[r_{P^*}^e]$ and σ_{P^*} solely relate to the original levels of P scaled by the same factor w_P. It follows that the Sharpe ratio is unaffected by the allocation between P and the risk-free asset, and formalizes the initial "linear function" understanding.

By imposing the matching volatility constraint we can derive the required weight in portfolio P:

$$\begin{aligned} & \sigma_{P^*} = w_P\, \sigma_P = \sigma_M \\ \implies \qquad & w_P = \frac{\sigma_M}{\sigma_P} = \frac{0.25}{0.35} = \frac{5}{7} \approx 0.7143 \\ \implies \qquad & w_f = 1 - w_P = \frac{2}{7} \approx 0.2857 \end{aligned}$$

Substituting into the excess-return expression above yields the expected excess return for P^*:

$$E[r^e_{P^*}] = w_P \, E[r^e_P] = \tfrac{5}{7} \cdot 15\% \approx 10.71\%$$

The portfolio P^* that matches the volatility of the market therefore invests 71.4% in P and 28.6% in the risk-free asset.

2.2 (b) Equivalence of the two comparisons

Following the results from (a), combining P with the risk-free asset scales the mean and volatility by the same factor, and P^* inherits P's Sharpe ratio:

$$S_{P^*} = \frac{E[r^e_{P^*}]}{\sigma_{P^*}} = \frac{w_P \, E[r^e_P]}{w_P \, \sigma_P} = \frac{E[r^e_P]}{\sigma_P} = S_P$$

The Sharpe ratio is independent of the allocation between P and the risk-free asset as the scaling term cancels out.

Using this fact, $S_P = S_{P^*}$, and the initial constraint $\sigma_{P^*} = \sigma_M$, both terms can be rewritten to have a common denominator:

$$S_P - S_M = S_{P^*} - S_M = \frac{E[r^e_{P^*}]}{\sigma_{P^*}} - \frac{E[r^e_M]}{\sigma_M} = \frac{E[r^e_{P^*}] - E[r^e_M]}{\sigma_M}$$

The two comparisons therefore rank P and M identically. This can also be numerically verified as the comparison yields

$$S_P - S_M = 0.4286 - 0.4000 = 0.0286, \qquad \frac{E[r^e_{P^*}] - E[r^e_M]}{\sigma_M} = \frac{10.71\% - 10.00\%}{25\%} = 0.0286.$$

Comparing $E[r^e_{P^*}]$ with $E[r^e_M]$ is the more intuitive of the two. It answers how much additional expected return the investor earns by choosing P over the market portfolio, while holding volatility fixed at the market's level of 25%. In this example, the investor earns 0.71 percentage points more per year by choosing P over M, which is economically concrete.

The Sharpe ratio comparison has the opposite advantage, that it is applied to the two initial portfolios, without the need to construct a synthetic portfolio P^*, and it is unaffected by leverage.

2.3 (c) Optimal combination of P and M

Portfolio P offers a higher Sharpe ratio than the market, and a market beta of 1.2. By using a single-factor model, we can separate P's excess return into alpha, market exposure, and residual risk with no expected excess return:

$$r^e_P = \alpha_P + \beta_P \, r^e_M + \varepsilon_P, \qquad E[\varepsilon_P] = 0, \qquad \mathrm{Cov}(\varepsilon_P, r^e_M) = 0$$

The alpha of P is the expected excess return not explained by its market exposure:

$$\alpha_P = E[r_P^e] - \beta_P\, E[r_M^e] = 0.15 - 1.2 \cdot 0.10 = 0.03 = 3\%$$

The total portfolio variance consists of a systematic component, scaled by the squared market beta, and an idiosyncratic component. Solving for the idiosyncratic part yields:

$$\sigma_\varepsilon^2 = \sigma_P^2 - \beta_P^2 \sigma_M^2 = 0.35^2 - 1.2^2 \cdot 0.25^2 = 0.1225 - 0.0900 = 0.0325,$$

$$\sigma_\varepsilon = \sqrt{0.0325} \approx 0.1803 = 18.03\%, \qquad \mathrm{Cov}(r_P, r_M) = \beta_P \sigma_M^2 = 1.2 \cdot 0.0625 = 0.075.$$

To find the optimal weight w_P^0 in the unadjusted active position, we compare its alpha per unit of residual variance with the market's reward per unit of market variance:

$$w_P^0 = \frac{\alpha_P/\sigma_\varepsilon^2}{E[r_M^e]/\sigma_M^2} = \frac{0.03/0.0325}{0.10/0.0625} = \frac{0.9231}{1.6000} = \frac{15}{26} \approx 0.5769.$$

Because P is not beta-neutral, a position in P also carries market exposure. We therefore adjust the weight by the portfolio's market beta

$$w_P = \frac{w_P^0}{1 + (1 - \beta_P)\, w_P^0} = \frac{15/26}{1 - 0.2 \cdot 15/26} = \frac{15/26}{23/26} = \frac{15}{23} \approx 0.6522$$

$$w_M = 1 - w_P = \frac{8}{23} \approx 0.3478.$$

The optimal risky portfolio P_{opt} therefore holds 65.2% in P and 34.8% in the market portfolio. Since $\beta_P > 1$, the denominator of the adjustment is smaller than one and $w_P > w_P^0$. The resulting market exposure of the combined portfolio is

$$\beta_{\text{opt}} = w_P \beta_P + w_M = \tfrac{15}{23} \cdot 1.2 + \tfrac{8}{23} \approx 1.13.$$

2.4 (d) Sharpe ratio of the optimal portfolio

The squared Sharpe ratio of the optimal portfolio consists of the squared Sharpe ratio of the market and the squared contribution of the active position $\alpha_P/\sigma_\varepsilon$, measured as alpha per unit of residual risk:

$$S_{\text{opt}}^2 = S_M^2 + \left(\frac{\alpha_P}{\sigma_\varepsilon}\right)^2$$

Substituting the respective values yields a Sharpe ratio

$$\frac{\alpha_P}{\sigma_\varepsilon} = \frac{0.03}{0.1803} \approx 0.1664$$

$$S_{\text{opt}}^2 = 0.4000^2 + 0.1664^2 = 0.1600 + 0.0277 = 0.1877 \qquad \Longrightarrow \qquad S_{\text{opt}} = \sqrt{0.1877} \approx 0.4332.$$

Alternatively, we can compute the excess return of P_{opt} and its variance to find its Sharpe ratio:

$$E[r_{\text{opt}}^e] = \tfrac{15}{23} \cdot 0.15 + \tfrac{8}{23} \cdot 0.10 = 0.1326$$

and

$$\sigma_{\text{opt}}^2 = w_P^2 \sigma_P^2 + w_M^2 \sigma_M^2 + 2 w_P w_M \operatorname{Cov}(r_P, r_M)$$

$$\sigma_{\text{opt}}^2 = \left(\tfrac{15}{23}\right)^2 (0.1225) + \left(\tfrac{8}{23}\right)^2 (0.0625) + 2\left(\tfrac{15}{23}\right)\left(\tfrac{8}{23}\right)(0.075) \approx 0.0937$$

$$\sigma_{\text{opt}} = \sqrt{0.0937} \approx 0.3061.$$

Calculating the Sharpe ratio now, arrives at the same level as before:

$$S_{\text{opt}} = \frac{E[r_{\text{opt}}^e]}{\sigma_{\text{opt}}} = \frac{0.1326}{0.3061} \approx 0.4332.$$

The optimal portfolio has a Sharpe ratio of 0.4332, which exceeds both $S_P \approx 0.4286$ and $S_M = 0.4000$, confirming that combining the two portfolios dominates holding either one in isolation.

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3 Numerical Exercise: Evaluating Mutual Fund Performance

We evaluate the performance of four US equity mutual funds over the period January 2002 to July 2026 (last monthly observation 2026-07-01, $T = 295$ months):

Ticker	Fund	Style suggested by the name
TRBCX	T. Rowe Price Blue Chip Growth	Large-cap growth
GLCGX	Goldman Sachs Large Cap Growth Insights	Large-cap growth
STSEX	BlackRock Exchange Portfolio	Large-cap blend
DFSVX	DFA U.S. Small Cap Value Portfolio	Small-cap value

We proceed in three steps:

1. Load monthly adjusted prices from Yahoo Finance and the risk-free rate and Fama-French-Carhart (FFC) four-factor returns from the Kenneth French data library, and compute monthly excess returns.

2. Compute the Sharpe ratio and the information ratio relative to the market for each fund.
3. Run factor regressions (FF3 and FFC4) to test whether there is evidence of fund manager skill, i.e. a statistically significant positive alpha.

Claude (Opus 5) was used to generate the Python code in this section and as a sounding board for the econometric specifications and robustness checks. All results were verified against the output, and the interpretation is our own.

First, ensure the required packages are available and set universal formats for the following data analysis and graphics.

[127]:
```
import sys, subprocess, importlib

def install_and_import(package, alias=None):
    try:
        module = importlib.import_module(package)
    except ImportError:
        print(f"{package} not found. Installing...")
        subprocess.check_call([sys.executable, "-m", "pip", "install", package])
        module = importlib.import_module(package)
    globals()[alias or package] = module

install_and_import("statsmodels")

import numpy as np
import pandas as pd
import statsmodels.api as sm
import matplotlib.pyplot as plt
from pathlib import Path
from scipy import stats
from statsmodels.iolib.summary2 import summary_col

pd.set_option("display.float_format", lambda x: "%.4f" % x)
pd.set_option("display.width", 110)

# Colours (fixed per fund throughout), market in neutral grey
FUNDS = ["TRBCX", "GLCGX", "STSEX", "DFSVX"]
COL = {"TRBCX": "#2a78d6", "GLCGX": "#eb6834", "STSEX": "#1baf7a", "DFSVX":␣
 ↪"#eda100", "MKT": "#52514e"}
NAMES = {"TRBCX": "T. Rowe Price Blue Chip Growth", "GLCGX": "GS Large Cap␣
 ↪Growth Insights",
         "STSEX": "BlackRock Exchange Portfolio", "DFSVX": "DFA US Small Cap␣
 ↪Value", "MKT": "Market"}
plt.rcParams.update({"axes.spines.top": False, "axes.spines.right": False,
                     "axes.grid": True, "grid.alpha": 0.3, "figure.dpi": 110})
```

[128]:
```
# Render pandas Styler tables in the LaTeX/PDF export as well as in the␣
 ↪notebook.
```

```
from pandas.io.formats.style import Styler

def _styler_to_latex(styler):
    body = styler.to_latex(convert_css=True, hrules=True).replace("%", r"\%")
    return r"\begin{adjustbox}{max width=\textwidth}" + "\n" + body + "\n" +␣
 ↪r"\end{adjustbox}"

get_ipython().display_formatter.formatters["text/latex"].for_type(Styler,␣
 ↪_styler_to_latex)
```

3.1 Data and monthly excess returns

For each fund we use the monthly *adjusted* closing price from Yahoo Finance. The adjusted price assumes that all dividends and capital gains distributions are reinvested, so the simple monthly return

$$r_{i,t} = \frac{P^{adj}_{i,t}}{P^{adj}_{i,t-1}} - 1$$

is the fund's total return. Because a fund's net asset value is reported after the deduction of the management fee and other expenses, these are net-of-fee returns, i.e. the returns that an investor in the fund actually earned. The price series starts in December 2001 so that the first return is the one for January 2002. Yahoo labels each monthly bar with the first day of the month (e.g. 2026-07-01), while the price itself is the closing price on the last trading day of that month, so the bar dated 2026-07-01 delivers the July 2026 return.

From the Kenneth French data library we use the monthly file *Fama/French 3 Factors* and the *Momentum Factor*, for the factors and risk-free rate:

- $r_{f,t}$ (RF): one-month Treasury bill return,
- $r_{M,t} - r_{f,t}$ (Mkt-RF): excess return of the value-weighted market portfolio of all CRSP firms listed on the NYSE, AMEX and NASDAQ,
- SMB_t (Small Minus Big): return on small-cap stocks minus large-cap stocks (size factor),
- HML_t (High Minus Low): return on high book-to-market (value) stocks minus low book-to-market (growth) stocks (value factor),
- Mom_t: return on stocks with high prior 2-12 month returns minus stocks with low prior returns (momentum factor of Carhart, 1997).

The file reports returns in percent, so all values are divided by 100. SMB, HML and Mom are returns on zero-investment long-short portfolios and are therefore already "excess" returns.

After matching both data sets by calendar month, the monthly excess return of fund i is

$$r^e_{i,t} = r_{i,t} - r_{f,t}.$$

The code below downloads the fund prices from Yahoo Finance and the factors from the Kenneth French data library and saves them as `fund_prices_yahoo.csv` and `ffc4_factors_french.csv` next to the notebook. If the two files already exist, they are loaded instead of downloaded again. The results below are based on data downloaded on 13 October 2026.

[129]:
```
DATA_DIR = Path.cwd()
PRICE_FILE = DATA_DIR / "fund_prices_yahoo.csv"
FACTOR_FILE = DATA_DIR / "ffc4_factors_french.csv"
FRENCH_URL = "https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/"

PRICE_START, PRICE_END = "2001-12-01", "2026-08-01"            # Yahoo end date␣
 ↪is exclusive
SAMPLE_START, SAMPLE_END = pd.Period("2002-01", "M"), pd.Period("2026-07", "M")


def download_prices(tickers, start, end):
    """Monthly adjusted closing prices from Yahoo Finance."""
    install_and_import("yfinance", "yf")
    raw = yf.download(tickers, start=start, end=end, interval="1mo",
                      auto_adjust=False, actions=False, progress=False)
    px = raw["Adj Close"][tickers].dropna(how="all")
    px.index = pd.to_datetime(px.index).to_period("M")          # 2026-07-01 ->␣
 ↪2026-07
    px.index.name = "date"
    return px


def download_french(zip_name):
    """Monthly table (in percent) from a zipped CSV file of the Kenneth French␣
 ↪data library."""
    import io, zipfile, requests
    resp = requests.get(FRENCH_URL + zip_name, timeout=60)
    resp.raise_for_status()
    with zipfile.ZipFile(io.BytesIO(resp.content)) as z:
        text = z.read(z.namelist()[0]).decode("latin-1")
    header, rows = None, []
    for line in text.splitlines():
        parts = [s.strip() for s in line.split(",")]
        if header is None:                                      # first line␣
 ↪like ",Mkt-RF,SMB,HML,RF"
            if len(parts) > 1 and parts[0] == "" and parts[1] != "":
                header = parts[1:]
            continue
        if len(parts) > 1 and parts[0].isdigit() and len(parts[0]) == 6:
            rows.append(parts)                                  # monthly row,␣
 ↪e.g. "200201,-1.44,..."
        elif rows:                                              # end of the␣
 ↪monthly block (annual data follow)
            break
    df = pd.DataFrame([r[1:] for r in rows], index=[r[0] for r in rows],␣
 ↪columns=header).astype(float)
```

```
    df.index.name = "date"
    return df


# ---- fund prices ----
if PRICE_FILE.exists():
    prices = pd.read_csv(PRICE_FILE, index_col="date")
    prices.index = pd.PeriodIndex(prices.index, freq="M")
    print(f"Fund prices loaded from local copy: {PRICE_FILE.name}")
else:
    prices = download_prices(FUNDS, PRICE_START, PRICE_END)
    prices.to_csv(PRICE_FILE, index_label="date")
    print("Fund prices downloaded from Yahoo Finance and saved.")

# ---- factors (in percent, as in the French library) ----
if FACTOR_FILE.exists():
    factors_pct = pd.read_csv(FACTOR_FILE, index_col="date")
    print(f"Factors loaded from local copy: {FACTOR_FILE.name}")
else:
    ff3 = download_french("F-F_Research_Data_Factors_CSV.zip")
    mom = download_french("F-F_Momentum_Factor_CSV.zip")
    factors_pct = ff3.join(mom, how="inner")
    factors_pct.to_csv(FACTOR_FILE, index_label="date")
    print("Factors downloaded from the Kenneth French data library and saved.")

factors_pct.index = pd.to_datetime(factors_pct.index.astype(str),␣
 ↪format="%Y%m").to_period("M")
factors_pct.columns = [c.strip() for c in factors_pct.columns]

prices = prices.loc[SAMPLE_START - 1:SAMPLE_END, FUNDS]          # December␣
 ↪2001 to July 2026
factors = factors_pct.loc[SAMPLE_START:SAMPLE_END, ["Mkt-RF", "SMB", "HML",␣
 ↪"Mom", "RF"]] / 100
prices.tail(3)
```

```
Fund prices loaded from local copy: fund_prices_yahoo.csv
Factors loaded from local copy: ffc4_factors_french.csv
```

[129]:

date	TRBCX	GLCGX	STSEX	DFSVX
2026-05	222.2700	34.6800	2681.5806	58.8568
2026-06	211.4700	33.3400	2642.7783	60.5997
2026-07	206.4600	31.7800	2841.8401	61.9858

[130]:
```
rets = prices.pct_change().dropna()                              # simple␣
 ↪monthly total returns
assert rets.index.equals(factors.index), "Fund and factor months do not line up"
```

```
assert rets.notna().all().all() and factors.notna().all().all(), "Missing␣
 ↪observations"

rf = factors["RF"]
excess = rets.sub(rf, axis=0)                                    # fund excess␣
 ↪returns
mkt = factors["Mkt-RF"]                                          # market␣
 ↪excess return

print(f"Sample: {excess.index[0].strftime('%b %Y')} to {excess.index[-1].
 ↪strftime('%b %Y')}, T = {len(excess)} months "
      f"(identical for all four funds and the factors)")
excess.head()
```

```
Sample: Jan 2002 to Jul 2026, T = 295 months (identical for all four funds and
the factors)
```

[130]:

date	TRBCX	GLCGX	STSEX	DFSVX
2002-01	-0.0304	-0.0241	-0.0107	0.0140
2002-02	-0.0305	-0.0384	0.0040	-0.0029
2002-03	0.0459	0.0247	0.0279	0.0973
2002-04	-0.0634	-0.0635	-0.0578	0.0438
2002-05	-0.0182	-0.0265	-0.0210	-0.0329

The table below summarises the monthly excess returns. Means are annualised by multiplying by 12 and standard deviations by multiplying by $\sqrt{12}$ (i.i.d. monthly returns, as in Assignment 2). For comparison we also report the annualised geometric total return, $\left(\prod_t (1 + r_{i,t})\right)^{12/T} - 1$, i.e. the realised compound growth rate of an investment in the fund. The market's total return is $(r_{M,t} - r_{f,t}) + r_{f,t}$.

[131]:
```
ex_all = excess.assign(MKT=mkt)                                  # funds +␣
 ↪market, excess returns
tot_all = rets.assign(MKT=mkt + rf)                              # funds +␣
 ↪market, total returns
T = len(ex_all)

summary = pd.DataFrame({
    "Mean excess return (ann.)": ex_all.mean() * 12,
    "SD (ann.)": ex_all.std() * np.sqrt(12),
    "Min (monthly)": ex_all.min(),
    "Max (monthly)": ex_all.max(),
    "Correlation with market": ex_all.corr()["MKT"],
    "Geometric total return (ann.)": (1 + tot_all).prod() ** (12 / T) - 1,
}).rename(index=NAMES)
print(f"Factor premia (ann.):  Mkt-RF {mkt.mean()*12:.2%}   "
      f"SMB {factors['SMB'].mean()*12:.2%}   "
```

```
      f"HML {factors['HML'].mean()*12:.2%}    "
      f"Mom {factors['Mom'].mean()*12:.2%}")
summary.style.format({c: "{:.2%}" for c in summary.columns if c != "Correlation␣
 ↪with market"}
                     | {"Correlation with market": "{:.3f}"})
```

Factor premia (ann.): Mkt-RF 9.31% SMB 0.74% HML 0.70% Mom 2.03%

[131]:

	Mean excess return (ann.)	SD (ann.)	Min (monthly)	Max (monthly)	Correlation with market	Geometric total return (ann.)
T. Rowe Price Blue Chip Growth	9.85%	17.14%	-18.85%	14.99%	0.937	10.51%
GS Large Cap Growth Insights	8.53%	16.35%	-17.68%	15.40%	0.962	9.22%
BlackRock Exchange Portfolio	7.61%	13.27%	-12.26%	10.79%	0.925	8.73%
DFA US Small Cap Value	10.71%	21.47%	-26.71%	19.65%	0.864	10.53%
Market	9.31%	15.27%	-17.20%	13.60%	1.000	10.26%

Relative to the market, which earned an average excess return of 9.31% p.a. with a volatility of 15.27%, the four funds fall into three groups:

- **Higher return and higher volatility: TRBCX and DFSVX.** TRBCX earned 0.54 percentage points (pp) p.a. more than the market (9.85%), but with 1.87 pp more volatility (17.14%). DFSVX earned the highest excess return of all funds, 1.40 pp above the market (10.71%), but at the cost of by far the highest volatility, 6.20 pp above the market (21.47%), and and its own worst month of the entire sample (an excess return of -26.7% in March 2020, a month in which the market lost 13.4%). For both funds the additional return is small relative to the additional risk, which already indicates that neither offers a better risk-return trade-off than the market (see Section 2).
- **Lower return and higher volatility: GLCGX.** GLCGX earned 0.78 pp p.a. less than the market (8.53%) while being more volatile (16.35%). In terms of mean and standard deviation, the market dominated this fund over the sample.
- **Lower return and lower volatility: STSEX.** STSEX earned 1.70 pp p.a. less than the market (7.61%), but was also clearly less volatile (13.27%) and had the mildest worst month of the four (-12.3%, against the market's own worst month of -17.2%). Whether the lower return is adequate compensation for the lower risk can only be judged on a risk-adjusted basis.

Correlations with the market are high (0.86 to 0.96), as expected for diversified equity funds. The lowest correlation belongs to DFSVX, whose small-cap value holdings differ most from the value-weighted market, which is dominated by large firms.

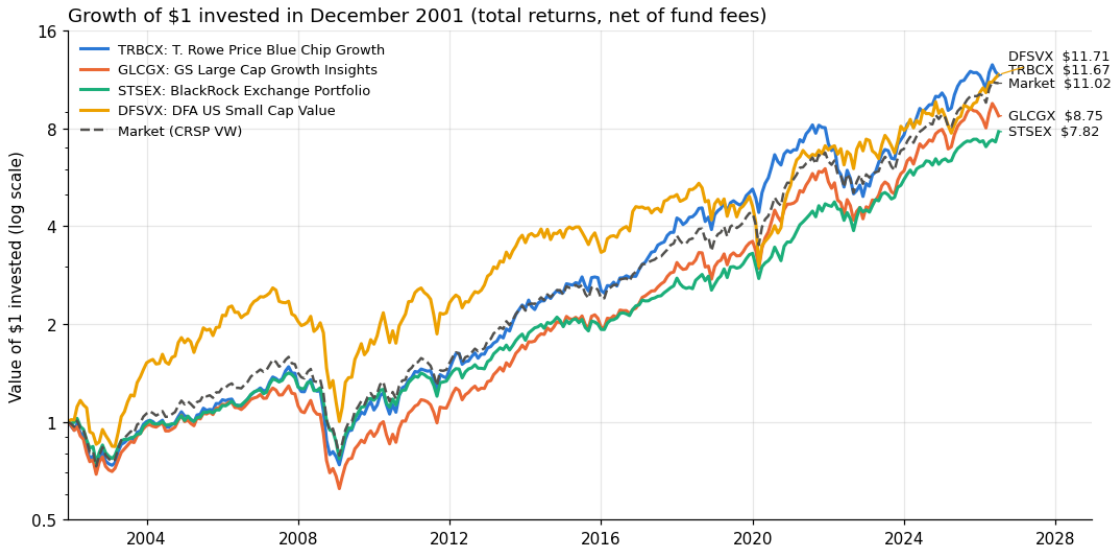
The chart below shows the growth of USD 1 invested in each fund and in the market in December 2001 (log scale, so that equal slopes correspond to equal percentage returns).

[132]:

```
growth = (1 + tot_all).cumprod()
growth.loc[SAMPLE_START - 1] = 1.0
growth = growth.sort_index()
x = growth.index.to_timestamp()

fig, ax = plt.subplots(figsize=(10, 5))
for k in FUNDS + ["MKT"]:
    ax.plot(x, growth[k], color=COL[k], lw=2 if k != "MKT" else 1.6,
            ls="--" if k == "MKT" else "-", label=f"{k}: {NAMES[k]}" if k !=␣
 ↪"MKT" else "Market (CRSP VW)")
```

```
# end-of-period labels, spread out vertically so that they do not overlap
final = growth.iloc[-1].sort_values()
ypos, prev = {}, None
for k, v in final.items():
    y = v if prev is None else max(v, prev * 1.10)
    ypos[k], prev = y, y
for k, v in final.items():
    ax.annotate(f"{k if k != 'MKT' else 'Market'}  ${v:.2f}", (x[-1], v),␣
 ↪xytext=(x[-1] + pd.Timedelta(days=90), ypos[k]),
                va="center", fontsize=8.5, color="#0b0b0b",
                arrowprops=dict(arrowstyle="-", color=COL[k], lw=0.8))
ax.set_yscale("log")
ax.set_yticks([0.5, 1, 2, 4, 8, 16]); ax.set_yticklabels(["0.5", "1", "2", "4",␣
 ↪"8", "16"])
ax.set_ylabel("Value of $1 invested (log scale)")
ax.set_title("Growth of $1 invested in December 2001 (total returns, net of␣
 ↪fund fees)", loc="left")
ax.legend(loc="upper left", frameon=False, fontsize=8.5)
ax.set_xlim(x[0], x[-1] + pd.Timedelta(days=900))
plt.tight_layout(); plt.show()
```



Over the full period, an investment of USD 1 in TRBCX or DFSVX grew to slightly more than the same investment in the market (USD 11.67 and USD 11.71, against USD 11.02), while GLCGX and STSEX ended clearly below the market. The paths are very different, however. DFSVX lost about twice as much as the market in March 2020 and went sideways from 2018 to 2020, a period of very poor performance for small-cap value stocks, while the growth funds benefited from the strong performance of large technology stocks after 2015 and fell sharply in 2022. Raw returns therefore mix differences in risk with differences in performance, which is why we turn to risk-adjusted measures next.

3.2 Sharpe ratio and information ratio

The Sharpe ratio measures the expected excess return per unit of total risk,

$$\hat{S}_i = \frac{\bar{r}_i^e}{\hat{\sigma}_i},$$

where $\bar{r}_i^e$ and $\hat{\sigma}_i$ are the sample mean and standard deviation of the monthly excess returns. It is the slope of the CAL obtained by combining the fund with T-bills, so it is the relevant measure if the fund is an investor's *entire* risky portfolio. With i.i.d. returns the mean scales with 12 and the standard deviation with $\sqrt{12}$, so $\hat{S}_i^{ann} = \sqrt{12}\,\hat{S}_i$.

The information ratio measures performance relative to a benchmark, here the market. The active return is the difference between the fund return and the benchmark return, in which the risk-free rate cancels out,

$$a_{i,t} = r_{i,t} - r_{M,t} = r_{i,t}^e - r_{M,t}^e, \qquad \widehat{IR}_i = \frac{\bar{a}_i}{sd(a_i)},$$

where the denominator $sd(a_i)$ is the **tracking error**. The IR is the average outperformance relative to the market per unit of risk taken by deviating from it, and is annualised with $\sqrt{12}$ as well. Since $sd(a_i)/\sqrt{T}$ is the standard error of the mean active return, the t-statistic for $H_0 : \mathbb{E}[a_i] = 0$ (no out- or underperformance of the market) is simply the monthly information ratio multiplied by $\sqrt{T}$,

$$t_{\bar{a}_i} = \frac{\bar{a}_i}{sd(a_i)/\sqrt{T}} = \widehat{IR}_i\,\sqrt{T}.$$

Appraisal ratio (CAPM). The IR implicitly assumes that the fund has a market beta of one. As a complement, the regression-based appraisal ratio first removes the fund's market exposure,

$$r_{i,t}^e = \alpha_i + \beta_i\, r_{M,t}^e + \varepsilon_{i,t}, \qquad \widehat{AR}_i = \frac{\hat{\alpha}_i}{sd(\hat{\varepsilon}_i)}.$$

Averaging the regression over the sample (OLS residuals have mean zero) links the two measures exactly:

$$\bar{a}_i = \bar{r}_i^e - \bar{r}_M^e = \hat{\alpha}_i + (\hat{\beta}_i - 1)\,\bar{r}_M^e.$$

The average active return thus consists of CAPM alpha plus a part that only reflects a market exposure above or below one.

[133]:
```
active = excess.sub(mkt, axis=0)                                  # a_t = r_i - r_M
t_active = active.mean() / (active.std() / np.sqrt(len(active)))   # = monthly IR x sqrt(T)
p_active = pd.Series(2 * stats.t.sf(t_active.abs(), df=len(active) - 1), index=FUNDS)
capm = {k: sm.OLS(excess[k], sm.add_constant(mkt)).fit() for k in FUNDS}

perf = pd.DataFrame({
    "Mean excess return (ann.)": excess.mean() * 12,
    "SD (ann.)": excess.std() * np.sqrt(12),
```

```
    "Sharpe ratio (monthly)": excess.mean() / excess.std(),
    "Sharpe ratio (ann.)": excess.mean() / excess.std() * np.sqrt(12),
    "Mean active return (ann.)": active.mean() * 12,
    "Tracking error (ann.)": active.std() * np.sqrt(12),
    "Information ratio (monthly)": active.mean() / active.std(),
    "Information ratio (ann.)": active.mean() / active.std() * np.sqrt(12),
    "t-stat mean active return": t_active,
    "p-value mean active return": p_active,
    "CAPM beta": pd.Series({k: capm[k].params["Mkt-RF"] for k in FUNDS}),
    "CAPM R-squared": pd.Series({k: capm[k].rsquared for k in FUNDS}),
    "CAPM alpha (ann.)": pd.Series({k: capm[k].params["const"] * 12 for k in␣
 ↪FUNDS}),
    "Beta effect, (beta - 1) x mean mkt excess (ann.)": pd.Series({k: (capm[k].
 ↪params["Mkt-RF"] - 1) * mkt.mean() * 12 for k in FUNDS}),
    "Appraisal ratio (ann.)": pd.Series({k: capm[k].params["const"] / np.
 ↪sqrt(capm[k].mse_resid) * np.sqrt(12) for k in FUNDS}),
})
perf.loc["MKT"] = [mkt.mean() * 12, mkt.std() * np.sqrt(12), mkt.mean() / mkt.
 ↪std(), mkt.mean() / mkt.std() * np.sqrt(12),
                   0, 0, np.nan, np.nan, np.nan, np.nan, 1, 1, 0, 0, np.nan]

pct_cols = ["Mean excess return (ann.)", "SD (ann.)", "Mean active return (ann.
 ↪)", "Tracking error (ann.)",
            "CAPM alpha (ann.)", "Beta effect, (beta - 1) x mean mkt excess␣
 ↪(ann.)"]
perf.rename(index=NAMES).T.style.format(
    lambda v: "" if pd.isna(v) else f"{v:.3f}"
).format(lambda v: f"{v:.2%}", subset=pd.IndexSlice[pct_cols, :])
```

[133]:

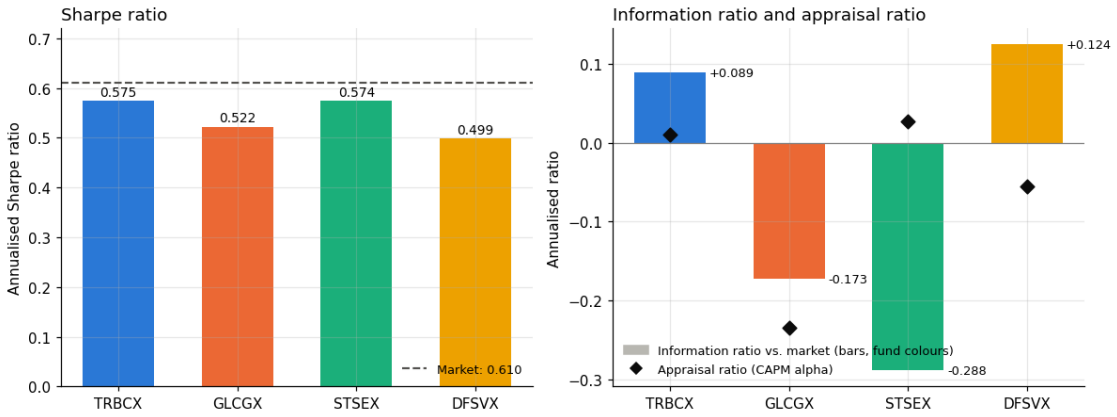
	T. Rowe Price Blue Chip Growth	GS Large Cap Growth Insights	BlackRock Exchange Portfolio	DFA US Small Cap Value	Market
Mean excess return (ann.)	9.85%	8.53%	7.61%	10.71%	9.31%
SD (ann.)	17.14%	16.35%	13.27%	21.47%	15.27%
Sharpe ratio (monthly)	0.166	0.151	0.166	0.144	0.176
Sharpe ratio (ann.)	0.575	0.522	0.574	0.499	0.610
Mean active return (ann.)	0.54%	-0.78%	-1.70%	1.40%	0.00%
Tracking error (ann.)	6.05%	4.50%	5.88%	11.28%	0.00%
Information ratio (monthly)	0.026	-0.050	-0.083	0.036	
Information ratio (ann.)	0.089	-0.173	-0.288	0.124	
t-stat mean active return	0.441	-0.857	-1.429	0.617	
p-value mean active return	0.660	0.392	0.154	0.538	
CAPM beta	1.051	1.029	0.803	1.215	1.000
CAPM R-squared	0.878	0.925	0.855	0.747	1.000
CAPM alpha (ann.)	0.06%	-1.05%	0.14%	-0.60%	0.00%
Beta effect, (beta - 1) x mean mkt excess (ann.)	0.48%	0.27%	-1.83%	2.00%	0.00%
Appraisal ratio (ann.)	0.010	-0.234	0.027	-0.056	

[134]:
```
fig, axes = plt.subplots(1, 2, figsize=(11, 4.2))
xx = np.arange(len(FUNDS))

ax = axes[0]
sr = perf.loc[FUNDS, "Sharpe ratio (ann.)"]
ax.bar(xx, sr, color=[COL[k] for k in FUNDS], width=0.6)
ax.axhline(perf.loc["MKT", "Sharpe ratio (ann.)"], color=COL["MKT"], ls="--",␣
 ↪lw=1.4,
           label=f"Market: {perf.loc['MKT', 'Sharpe ratio (ann.)']:.3f}")
```

```
for i, v in enumerate(sr):
    ax.text(i, v + 0.01, f"{v:.3f}", ha="center", fontsize=9)
ax.set_xticks(xx, FUNDS); ax.set_ylabel("Annualised Sharpe ratio"); ax.
 ↪set_ylim(0, 0.72)
ax.set_title("Sharpe ratio", loc="left"); ax.legend(frameon=False, fontsize=8.
 ↪5, loc="lower right")

ax = axes[1]
ir = perf.loc[FUNDS, "Information ratio (ann.)"]
ar = perf.loc[FUNDS, "Appraisal ratio (ann.)"]
ax.bar(xx, ir, color=[COL[k] for k in FUNDS], width=0.6)
ax.scatter(xx, ar, marker="D", s=46, color="#0b0b0b", zorder=3,␣
 ↪label="Appraisal ratio (CAPM alpha)")
ax.axhline(0, color="grey", lw=0.8)
for i, v in enumerate(ir):
    ax.text(i + 0.33, v, f"{v:+.3f}", ha="left", va="center", fontsize=8.5)
ax.set_xticks(xx, FUNDS); ax.set_ylabel("Annualised ratio")
ax.set_title("Information ratio and appraisal ratio", loc="left")
from matplotlib.patches import Patch
from matplotlib.lines import Line2D
ax.legend(handles=[Patch(color="#b8b7b1", label="Information ratio vs. market␣
 ↪(bars, fund colours)"),
                   Line2D([], [], marker="D", ls="", color="#0b0b0b",␣
 ↪label="Appraisal ratio (CAPM alpha)")],
          frameon=False, fontsize=8.5, loc="lower left")
plt.tight_layout(); plt.show()
```



3.2.1 Discussion

Sharpe ratios: None of the four funds achieved a higher Sharpe ratio than the market (0.610). TRBCX (0.575) and STSEX (0.574) come closest, followed by GLCGX (0.522) and DFSVX (0.499). An investor who held the market portfolio and T-bills, choosing the mix that gives the desired level

of risk on the market's CAL, would therefore have obtained a better risk-return trade-off than with any one of these funds as the entire risky portfolio. It is interesting that TRBCX and STSEX reach almost the same Sharpe ratio in very different ways: TRBCX earns a high average excess return (9.85%) with high volatility (17.1%), whereas STSEX earns less (7.61%) with much lower volatility (13.3%). DFSVX has the highest average excess return of all (10.71%), but its volatility of 21.5% is so high that it ends up with the lowest Sharpe ratio.

Information ratios: Relative to the market, TRBCX (IR = 0.089) and DFSVX (IR = 0.124) outperformed on average by 0.54% and 1.40% p.a., with tracking errors of 6.1% and 11.3%. GLCGX (IR = -0.173) and STSEX (IR = -0.288) underperformed by 0.78% and 1.70% p.a. All information ratios are small. The t-statistics of the mean active returns reported in the table range from -1.43 (STSEX, $p = 0.15$) to 0.61 (DFSVX), so none of the funds out- or underperformed the market significantly.

Evidence of skill: The decomposition $\bar{a}_i = \hat{\alpha}_i + (\hat{\beta}_i - 1)\bar{r}^e_M$ shows that most of the active returns are explained by market beta and not by alpha. DFSVX has a beta of 1.22, so its higher market exposure alone accounts for +2.00% p.a. of active return, which more than explains the +1.40% it actually earned; its CAPM alpha is -0.60%. Conversely, STSEX's underperformance of 1.70% p.a. comes entirely from its low beta of 0.80 (-1.83% p.a.), while its CAPM alpha is slightly positive (+0.14%). In a period in which the market earned a high premium (9.31% p.a.), any fund with $\beta > 1$ looks good and any fund with $\beta < 1$ looks bad against the market, even though an investor could create the same market exposure with leverage or T-bills at no cost. The appraisal ratios, which correct for beta, are all close to zero (between -0.23 for GLCGX and +0.03 for STSEX). Whether the remaining small alphas reflect skill or only exposure to other systematic factors such as size and value is what we test in the next section.

3.3 Evidence of fund manager skill: factor regressions

A manager has skill if the fund earns a higher return than an investor could obtain with a passive portfolio that has the same systematic risk exposures. In a factor model, this abnormal return is measured by the intercept α_i of a time-series regression of the fund's excess return on the factor returns. Since the right-hand side variables are returns on traded (excess-return) portfolios, $\sum_k \beta_{i,k} f_{k,t}$ is the return of a passive replicating portfolio of the factors and T-bills, and α_i is the average return of the fund in excess of that replicating portfolio. If the factor model correctly describes expected returns, $\alpha_i = 0$ for a fund without skill.

Fama-French three-factor model (FF3):

$$r^e_{i,t} = \alpha_i + \beta_{i,M}\,(r_{M,t} - r_{f,t}) + \beta_{i,S}\,SMB_t + \beta_{i,H}\,HML_t + \varepsilon_{i,t}$$

Fama-French-Carhart four-factor model (FFC4):

$$r^e_{i,t} = \alpha_i + \beta_{i,M}\,(r_{M,t} - r_{f,t}) + \beta_{i,S}\,SMB_t + \beta_{i,H}\,HML_t + \beta_{i,Mom}\,Mom_t + \varepsilon_{i,t}$$

The factor loadings describe the fund's investment style: $\beta_{i,S} > 0$ indicates a tilt towards small stocks, $\beta_{i,H} > 0$ a tilt towards value stocks and $\beta_{i,H} < 0$ towards growth stocks, $\beta_{i,Mom} > 0$ a tilt towards past winners. A manager who simply follows a size, value or momentum style earns the corresponding factor premia, but this is not skill in the sense of the models because the same exposure can be obtained passively.

The following details the hypothesis test. A positive estimated alpha alone is not enough, since it could be the result of luck. For each fund we test

$$H_0 : \alpha_i = 0 \quad \text{against} \quad H_1 : \alpha_i \neq 0, \qquad t_{\alpha_i} = \frac{\hat{\alpha}_i}{SE(\hat{\alpha}_i)},$$

and reject H_0 at the conventional two-sided 5% level if $|t| > 1.97$ ($p < 0.05$). Evidence of skill requires a significantly *positive* alpha. Our main results use conventional OLS standard errors. Because monthly fund returns may show heteroskedasticity and some autocorrelation, we also report the p-value based on Newey-West (HAC) standard errors with 6 lags as a robustness check. The usual $4(T/100)^{2/9}$ rule of thumb gives five lags for $T = 295$, so this choice is slightly conservative. Alphas are monthly in the regressions and annualised by multiplying by 12. To see how alpha changes as factors are added, we also estimate the CAPM regression from Section 2 as a baseline.

[135]:
```
MODELS = {"CAPM": ["Mkt-RF"], "FF3": ["Mkt-RF", "SMB", "HML"], "FFC4":␣
 ↪["Mkt-RF", "SMB", "HML", "Mom"]}

res, res_hac = {}, {}
for m, cols in MODELS.items():
    X = sm.add_constant(factors[cols])
    for k in FUNDS:
        res[m, k] = sm.OLS(excess[k], X).fit()
        res_hac[m, k] = sm.OLS(excess[k], X).fit(cov_type="HAC",␣
 ↪cov_kwds={"maxlags": 6})


def alpha_table(model):
    rows = {}
    for k in FUNDS:
        r, h = res[model, k], res_hac[model, k]
        lo, hi = r.conf_int().loc["const"] * 12
        rows[k] = {
            "alpha (monthly)": r.params["const"],
            "alpha (ann.)": r.params["const"] * 12,
            "SE alpha (ann.)": r.bse["const"] * 12,
            "95% CI alpha (ann.)": f"[{lo:.2%}, {hi:.2%}]",
            "t(alpha)": r.tvalues["const"],
            "p-value": r.pvalues["const"],
            "p-value (Newey-West)": h.pvalues["const"],
            **{f"beta {c}": r.params[c] for c in MODELS[model]},
            **{f"t({c})": r.tvalues[c] for c in MODELS[model]},
            "R-squared": r.rsquared,
            "Adj. R-squared": r.rsquared_adj,
            "Residual SD (ann.)": np.sqrt(r.mse_resid * 12),
            "Appraisal ratio (ann.)": r.params["const"] / np.sqrt(r.mse_resid)␣
 ↪* np.sqrt(12),
        }
    return pd.DataFrame(rows)
```

```
def show(tab):
    pct = ["alpha (ann.)", "SE alpha (ann.)", "Residual SD (ann.)"]
    return tab.rename(columns=NAMES).style.format(
        lambda v: v if isinstance(v, str) else f"{v:.4f}"
    ).format(lambda v: f"{v:.2%}", subset=pd.IndexSlice[pct, :]
    ).format(lambda v: f"{v:.3%}", subset=pd.IndexSlice[["alpha (monthly)"], :])


def regression_output(model):
    out = summary_col([res[model, k] for k in FUNDS], model_names=FUNDS,␣
 ↪stars=True,
                      float_format="%.5f", info_dict={"N": lambda r: f"{int(r.
 ↪nobs)}"},
                      regressor_order=["const"] + MODELS[model])
    print(f"{model} regressions of monthly fund excess returns (standard errors␣
 ↪in parentheses)")
    print(out)
```

3.3.1 FF3 regressions

[136]:
```
regression_output("FF3")
```

```
FF3 regressions of monthly fund excess returns (standard errors in parentheses)

============================================================
                  TRBCX       GLCGX       STSEX      DFSVX
------------------------------------------------------------
const           0.00003     -0.00090    -0.00011    0.00005
                (0.00072)   (0.00057)   (0.00075)   (0.00059)
Mkt-RF          1.08970***  1.05814***  0.84726***  1.02871***
                (0.01685)   (0.01345)   (0.01768)   (0.01384)
SMB             -0.08109*** -0.06822*** -0.28865*** 0.84758***
                (0.02957)   (0.02360)   (0.03102)   (0.02429)
HML             -0.39542*** -0.27228*** 0.09951***  0.63635***
                (0.02329)   (0.01859)   (0.02443)   (0.01913)
R-squared       0.94059     0.95841     0.89096     0.97446
R-squared Adj.  0.93998     0.95798     0.88984     0.97420
N               295         295         295         295
============================================================
Standard errors in parentheses.
* p<.1, ** p<.05, ***p<.01
```

[137]:
```
show(alpha_table("FF3"))
```

[137]:

	T. Rowe Price Blue Chip Growth	GS Large Cap Growth Insights	BlackRock Exchange Portfolio	DFA US Small Cap Value
alpha (monthly)	0.003%	-0.090%	-0.011%	0.005%
alpha (ann.)	0.04%	-1.08%	-0.13%	0.07%
SE alpha (ann.)	0.86%	0.69%	0.90%	0.71%
95% CI alpha (ann.)	[-1.65%, 1.73%]	[-2.43%, 0.27%]	[-1.91%, 1.65%]	[-1.33%, 1.46%]
t(alpha)	0.0453	-1.5705	-0.1437	0.0926
p-value	0.9639	0.1174	0.8858	0.9263
p-value (Newey-West)	0.9642	0.1129	0.8562	0.9240
beta Mkt-RF	1.0897	1.0581	0.8473	1.0287
beta SMB	-0.0811	-0.0682	-0.2886	0.8476
beta HML	-0.3954	-0.2723	0.0995	0.6364
t(Mkt-RF)	64.6694	78.6749	47.9291	74.3176
t(SMB)	-2.7422	-2.8903	-9.3038	34.8889
t(HML)	-16.9784	-14.6473	4.0728	33.2614
R-squared	0.9406	0.9584	0.8910	0.9745
Adj. R-squared	0.9400	0.9580	0.8898	0.9742
Residual SD (ann.)	4.20%	3.35%	4.40%	3.45%
Appraisal ratio (ann.)	0.0093	-0.3218	-0.0294	0.0190

For the factor loadings we find that the FF3 loadings match the funds' stated investment styles closely, and all are highly significant:

- TRBCX ($\beta_M = 1.09$, $\beta_S = -0.08$, $\beta_H = -0.40$) and GLCGX ($\beta_M = 1.06$, $\beta_S = -0.07$, $\beta_H = -0.27$) are large-cap growth funds: slightly more market risk than the market, a small tilt towards large firms and a strong negative loading on value, i.e. a tilt towards growth stocks.
- STSEX ($\beta_M = 0.85$, $\beta_S = -0.29$, $\beta_H = +0.10$) holds very large, somewhat value-oriented firms, with a below-average market beta.
- DFSVX ($\beta_M = 1.03$, $\beta_S = 0.85$, $\beta_H = 0.64$) has by far the largest loadings on both SMB and HML, as expected for a small-cap value fund.

Adding SMB and HML increases the explanatory power considerably compared with the CAPM, especially for DFSVX, whose R^2 rises from 0.75 to 0.97. With R^2 between 0.89 and 0.97, the three factors capture almost all of the funds' return variation, and the remaining residual (non-factor) volatility is only 3.3% to 4.4% p.a.

No fund has a significant FF3 alpha. The annualised estimates are +0.04% for TRBCX ($t = 0.05$), -1.08% for GLCGX ($t = -1.57$, $p = 0.12$), -0.13% for STSEX ($t = -0.14$) and +0.07% for DFSVX ($t = 0.09$). We cannot reject $H_0 : \alpha_i = 0$ at the 5% level (nor at the 10% level) for any fund, and the Newey-West p-values lead to the same conclusion. The only alpha estimate of economically meaningful size, GLCGX's, is *negative*.

Compared with the CAPM, the FF3 alpha of DFSVX increases from -0.60% to +0.07% p.a. Under the CAPM, part of the fund's small-cap value exposure is picked up by a high market beta of 1.22. Once size and value are controlled for directly, the market beta falls to 1.03 and DFSVX's performance is fully explained by its factor exposures. TRBCX's alpha is close to zero under both models.

3.3.2 FFC4 regressions

[138]:
```
regression_output("FFC4")
```

```
FFC4 regressions of monthly fund excess returns (standard errors in parentheses)

============================================================
                    TRBCX       GLCGX       STSEX       DFSVX
```

```
============================================================
const           -0.00003    -0.00106*   0.00010     0.00012
                (0.00072)   (0.00057)   (0.00075)   (0.00059)
Mkt-RF          1.09482***  1.07065***  0.83171***  1.02385***
                (0.01795)   (0.01418)   (0.01866)   (0.01474)
SMB             -0.08038*** -0.06648*** -0.29081*** 0.84690***
                (0.02960)   (0.02339)   (0.03078)   (0.02431)
HML             -0.39061*** -0.26053*** 0.08491***  0.63178***
                (0.02401)   (0.01897)   (0.02496)   (0.01971)
Mom             0.01442     0.03525**   -0.04379**  -0.01370
                (0.01733)   (0.01369)   (0.01802)   (0.01423)
R-squared       0.94073     0.95934     0.89314     0.97454
R-squared Adj.  0.93991     0.95878     0.89166     0.97419
N               295         295         295         295
============================================================
Standard errors in parentheses.
* p<.1, ** p<.05, ***p<.01
```

[139]:
```
show(alpha_table("FFC4"))
```

[139]:

	T. Rowe Price Blue Chip Growth	GS Large Cap Growth Insights	BlackRock Exchange Portfolio	DFA US Small Cap Value
alpha (monthly)	-0.003%	-0.106%	0.010%	0.012%
alpha (ann.)	-0.04%	-1.28%	0.12%	0.14%
SE alpha (ann.)	0.87%	0.68%	0.90%	0.71%
95% CI alpha (ann.)	[-1.75%, 1.66%]	[-2.62%, 0.07%]	[-1.66%, 1.89%]	[-1.26%, 1.54%]
t(alpha)	-0.0483	-1.8644	0.1285	0.2000
p-value	0.9615	0.0633	0.8978	0.8416
p-value (Newey-West)	0.9620	0.0663	0.8706	0.8348
beta Mkt-RF	1.0948	1.0707	0.8317	1.0238
beta SMB	-0.0804	-0.0665	-0.2908	0.8469
beta HML	-0.3906	-0.2605	0.0849	0.6318
beta Mom	0.0144	0.0353	-0.0438	-0.0137
t(Mkt-RF)	61.0027	75.5002	44.5688	69.4742
t(SMB)	-2.7155	-2.8422	-9.4483	34.8419
t(HML)	-16.2707	-13.7344	3.4017	32.0485
t(Mom)	0.8318	2.5745	-2.4299	-0.9628
R-squared	0.9407	0.9593	0.8931	0.9745
Adj. R-squared	0.9399	0.9588	0.8917	0.9742
Residual SD (ann.)	4.20%	3.32%	4.37%	3.45%
Appraisal ratio (ann.)	-0.0100	-0.3844	0.0265	0.0412

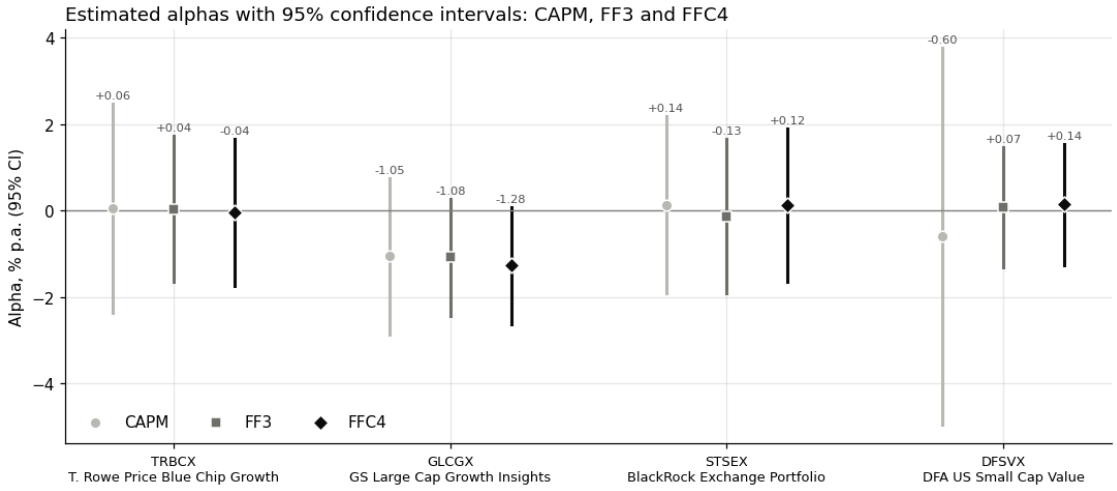
Momentum loadings: The loadings on the momentum factor are small for all four funds: +0.014 for TRBCX ($t = 0.8$), +0.035 for GLCGX ($t = 2.6$), -0.044 for STSEX ($t = -2.4$) and -0.014 for DFSVX ($t = -1.0$). Two of these loadings are statistically significant at the 5% level. GLCGX has a significant positive loading, which may come from a more systematic, algorithmic stock selection process, as the "Insights" in the fund's name suggests, if the models behind it tend to favour stocks with strong recent price trends. STSEX has a significant negative loading, which is consistent with a low-turnover portfolio that does not chase recent winners. For TRBCX and DFSVX, the momentum loadings are insignificant. Even where they are significant, the loadings are economically small. With a momentum premium of 2.03% p.a. over the sample, GLCGX's loading adds only about 0.07% p.a. to its expected return and STSEX's loading subtracts about 0.09% p.a. The other loadings and the R^2 are virtually unchanged from FF3 for all four funds, so none of them follows a momentum strategy in any meaningful sense.

Alphas: Adding momentum therefore changes the alphas only slightly: TRBCX -0.04% ($t = -0.05$), GLCGX -1.28% ($t = -1.86$, $p = 0.063$), STSEX +0.12% ($t = 0.13$) and DFSVX +0.14% ($t = 0.20$) p.a. Since the momentum factor earned a premium of 2.03% p.a. over the sample, GLCGX's

positive momentum exposure explains part of its return, so its alpha becomes more negative. Again, no fund has an alpha that is significantly different from zero at the 5% level. GLCGX's negative alpha is significant at the 10% level (Newey-West $p = 0.066$), which is weak evidence of *underperformance* and not of skill.

The chart below compares the annualised alpha estimates and their 95% confidence intervals across the three models.

[140]:
```
MCOL = {"CAPM": "#b8b7b1", "FF3": "#6f6e69", "FFC4": "#0b0b0b"}
MMARK = {"CAPM": "o", "FF3": "s", "FFC4": "D"}
fig, ax = plt.subplots(figsize=(10, 4.5))
offs = {"CAPM": -0.22, "FF3": 0.0, "FFC4": 0.22}
for m in MODELS:
    for i, k in enumerate(FUNDS):
        r = res[m, k]
        a = r.params["const"] * 12 * 100
        lo, hi = r.conf_int().loc["const"] * 12 * 100
        ax.plot([i + offs[m]] * 2, [lo, hi], color=MCOL[m], lw=2)
        ax.scatter(i + offs[m], a, color=MCOL[m], marker=MMARK[m], s=48,␣
 ↪zorder=3,
                   edgecolor="white", linewidth=1, label=m if i == 0 else None)
        ax.text(i + offs[m], hi + 0.12, f"{a:+.2f}", ha="center", fontsize=7.5,␣
 ↪color="#52514e")
ax.axhline(0, color="grey", lw=0.9)
ax.set_xticks(range(len(FUNDS)), [f"{k}\n{NAMES[k]}" for k in FUNDS],␣
 ↪fontsize=8.5)
ax.set_ylabel("Alpha, % p.a. (95% CI)")
ax.set_title("Estimated alphas with 95% confidence intervals: CAPM, FF3 and␣
 ↪FFC4", loc="left")
ax.legend(frameon=False, ncol=3, loc="lower left")
plt.tight_layout(); plt.show()
```



Next, we look at the robustness based on subperiods. A full-sample alpha of zero could hide skill in one part of the sample that disappears later, for example because a successful fund attracts inflows and grows past its capacity (Berk and Green, 2004). We therefore re-estimate the FFC4 regression separately over two subperiods of roughly equal length, 2002-01 to 2013-12 (144 months) and 2014-01 to 2026-07 (151 months).

[141]:
```
halves = {"2002-01 to 2013-12": (pd.Period("2002-01", "M"), pd.
 ↪Period("2013-12", "M")),
          "2014-01 to 2026-07": (pd.Period("2014-01", "M"), pd.
 ↪Period("2026-07", "M"))}
sub = {}
for lab, (s, e) in halves.items():
    X = sm.add_constant(factors.loc[s:e, MODELS["FFC4"]])
    for k in FUNDS:
        r = sm.OLS(excess.loc[s:e, k], X).fit()
        sub[k] = sub.get(k, {}) | {(lab, "alpha (ann.)"): r.params["const"] *␣
 ↪12,
                                   (lab, "t(alpha)"): r.tvalues["const"],
                                   (lab, "p-value"): r.pvalues["const"]}
sub = pd.DataFrame(sub).T
sub.columns = pd.MultiIndex.from_tuples(sub.columns, names=["FFC4, subperiod",␣
 ↪""])
sub.rename(index=NAMES).style.format("{:.3f}").format(
    "{:.2%}", subset=[c for c in sub.columns if c[1] == "alpha (ann.)"])
```

[141]:

FFC4, subperiod	2002-01 to 2013-12			2014-01 to 2026-07		
	alpha (ann.)	t(alpha)	p-value	alpha (ann.)	t(alpha)	p-value
T. Rowe Price Blue Chip Growth	0.22%	0.222	0.824	-0.57%	-0.411	0.682
GS Large Cap Growth Insights	-2.13%	-2.650	0.009	-0.61%	-0.543	0.588
BlackRock Exchange Portfolio	-0.21%	-0.209	0.835	0.40%	0.273	0.785
DFA US Small Cap Value	0.43%	0.497	0.620	-0.56%	-0.555	0.580

The subperiod results look the same as the full sample. TRBCX, STSEX and DFSVX have insignificant alphas with changing signs in both halves. GLCGX's underperformance is concentrated in the first half, where its FFC4 alpha of -2.13% p.a. is significant ($t = -2.65$); in the second half it is -0.61% p.a. and insignificant. There is no subperiod in which any fund delivers a significantly positive alpha.

3.3.3 Conclusion: is there evidence of fund manager skill?

No, neither the FF3 nor the FFC4 model provides evidence of manager skill for any of the four funds. All estimated alphas are statistically indistinguishable from zero at the 5% level, three of them are within $\pm 0.15\%$ p.a. of zero under both models, and the only economically sizeable alpha (GLCGX, about -1.1% to -1.3% p.a.) is negative and marginally significant at the 10% level under FFC4. The results are the same with Newey-West standard errors and in both halves of the sample, apart from GLCGX's significant underperformance before 2014.

The funds' performance is instead almost entirely explained by their exposures to the market,

size, value and momentum factors. This is consistent with the results from Section 2. The positive information ratios of TRBCX and DFSVX are the result of market betas above one and, for DFSVX, of its small-cap value style, not of superior stock selection. An investor could have replicated the risk and return of each fund with a passive portfolio of the factors, which is precisely the case when $\alpha = 0$.

There are several points to consider when interpreting these results:

- Net versus gross alpha: The fund returns are net of fees. An alpha of zero after fees implies a small positive alpha *before* fees, roughly equal to the annual expense ratio of the fund (a fraction of a percent per year for these funds). The managers may therefore have some ability to generate returns, but it is just enough to cover their fees, so that investors do not benefit from it. This is consistent with the "arithmetic of active management" (Sharpe, 1991): since passive investors hold the market portfolio, active investors in aggregate must hold it as well, so the average active fund earns a gross alpha of about zero and a negative net alpha after fees. It is also consistent with Berk and Green (2004), where fund flows compete away any net alpha.
- Statistical power: The standard errors of the annualised alphas are between 0.7% and 0.9%, so the 95% confidence intervals are about $\pm 1.3\%$ to $\pm 1.8\%$ p.a. wide. Even with almost 25 years of data, a true alpha of around $+1\%$ p.a., which would be economically very valuable, could not be distinguished from zero. The absence of evidence of skill is therefore not the same as evidence of the absence of skill, although the point estimates are centred close to zero.
- The factor benchmark is not free: SMB, HML and Mom are returns on long-short paper portfolios that ignore trading costs, short-selling costs and fees. For DFSVX, an alpha of zero relative to these factors means that the fund delivered its size and value exposure without losing return to costs, which still has value for investors who want this exposure. Measured against investable index funds instead of paper portfolios, its performance would probably look better.
- Selection and multiple testing: The four funds were chosen ex post among funds that survived the full 25-year period, which biases the sample towards successful funds. Moreover, with four funds and three model specifications, finding one alpha that is significant at the 10% level is not unusual even if all true alphas were zero. Both effects make it easier, not harder, to find positive alphas by chance, which strengthens the conclusion that there is no evidence of skill.

All in all, the regressions suggest that what investors pay for in these funds is not stock-picking skill but systematic factor exposure: market risk for all four funds, a growth tilt for TRBCX and GLCGX, a large-cap value tilt for STSEX and a small-cap value tilt for DFSVX.

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4 Bibliography

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